Stochastic Bandits

Tor Lattimore

June 8, 2026

Stochastic Bandits

A 1915 Mills Liberty Bell.

- → First half: The classical stuff (1950's-2010)
- → Second half: Brand new! (2026)

Why stochastic bandits?

- → Historical motivation: clinical trials [Thompson, 1933]. Actions are medical interventions (drugs). Rewards determined by patient responses. But there are many practical issues
- → Modern times: A/B testing, advertising

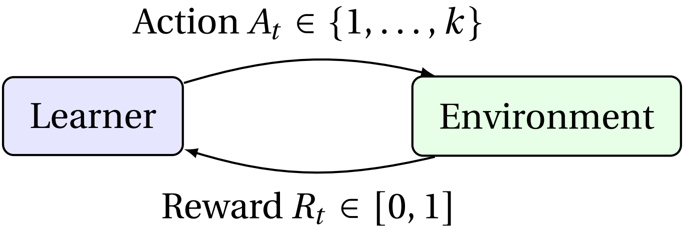
Why stochastic bandits *here*?

- $\rightarrow$ Bandits are single-state MDPs
- $\rightarrow$ The simplest model that demonstrates the exploration/exploitation dilemma
- $\rightarrow$ Simple models are a good place to design theoretically justified principled approaches

Formal model

- $\rightarrow$ k actions
- $\rightarrow$ *Unknown* mean vector $\mu \in [0, 1]^k$
- $\rightarrow$ *Known* time horizon n

Learner and environment interact over n rounds



- $\rightarrow$ Actions are $A_1, \ldots, A_n$
- $\rightarrow$ Rewards are $R_1, \ldots, R_n$
- $\rightarrow$ Assume the rewards are bounded in $[0, 1]$

Exploration/Exploitation dilemma

→ How do you decide between a known quantity and an unknown one

→ A problem I (and you) face *all the time*

- I like my favourite coffee shop. When a new one opens, do I try it?
- Should I bring my climbing shoes to Milan?
- Should I explore a new line of research?

Bandits and their many extensions capture the essence of this problem

Principles for exploration in bandits

- $\rightarrow$ Be greedy [This talk]
- $\rightarrow$ Explore naively [This talk]
- $\rightarrow$ Be optimistic [This talk]
- $\rightarrow$ Be Bayesian
- $\rightarrow$ Information-directed sampling
- $\rightarrow$ Reframe as optimisation [Nicoló?]

Performance measures

→ How do we assess the performance of a learning algorithm?

→ Convention: $\mu_1 > \mu_2 \geq \cdots \geq \mu_k$

→ The standard in the bandit literature is to **minimise** the **expected cumulative regret**

$$\mathrm{Reg}_n = \mathbb{E}\left[\sum_{t=1}^{n}(\mu_1 - \mu_{A_t})\right] = \mathbb{E}\left[\sum_{t=1}^{n}\Delta_{A_t}\right]$$

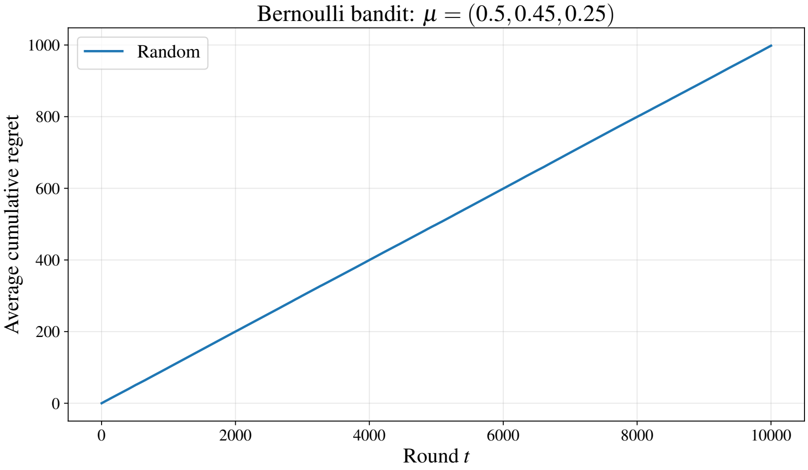
where $\Delta_a = \mu_1 - \mu_a$

→ Lot's of alternatives

Be greedy

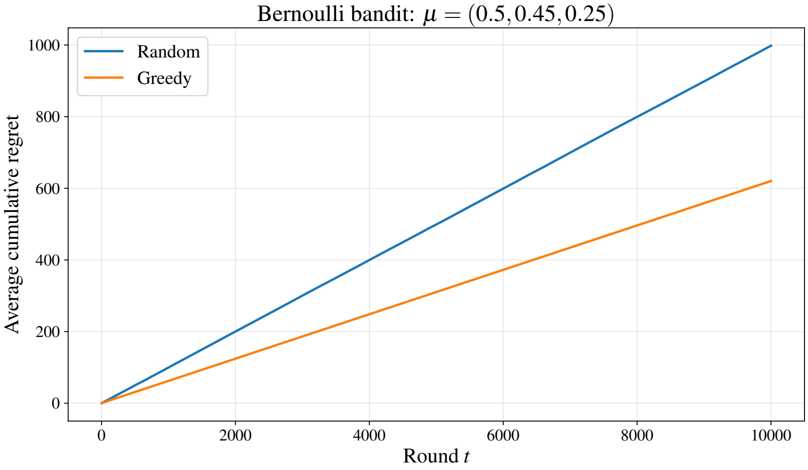
- → Play the action that currently looks best
- → Break ties randomly

Simulation



Bernoulli bandit: $\mu = (0.5, 0.45, 0.25)$
Random
Average cumulative regret
Round t
0
200
400
600
800
1000
0
2000
4000
6000
8000
10000

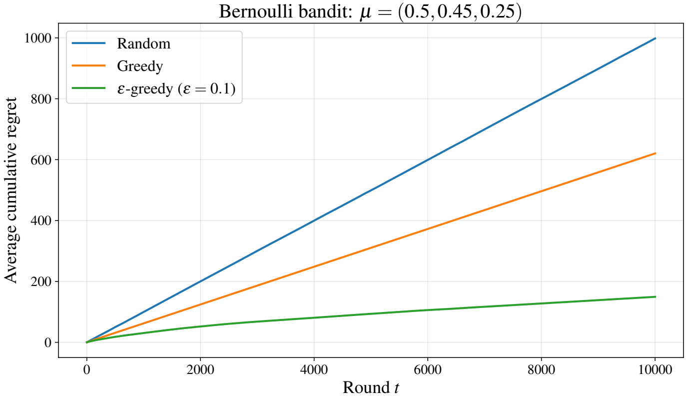
Simulation


Bernoulli bandit: $\mu = (0.5, 0.45, 0.25)$
Random
Greedy
Average cumulative regret
Round t
0
200
400
600
800
1000
0
2000
4000
6000
8000
10000

Explore naively

→ With probability $\epsilon \in [0, 1]$ play randomly

→ Else play the action that currently looks best



Optimism principle

*"Act as if you are in the nicest **plausibly possible** environment"*

Exploration and exploitation in bandits

- Optimism principle:

  "Act as if you are in the nicest ***plausibly possible*** *environment"*

- The action you should play in a bandit with means ν is $\arg\max_{a \in 1 \cdots k} \nu_a$
- The reward you would expect is $\max_{a \in 1 \cdots k} \nu_a$
- By the optimism

$$A_t = \arg\max_{a \in 1 \cdots k} \max_{\nu \in P_t} \nu_t$$

with P_t the set of **plausibly possible** means

Upper Confidence Bound Algorithm

A simple choice for the plausible set is

$$P_t = \left\{ \nu \in [0,1]^k : |\nu_a - \hat{\mu}_a(t-1)| \leq \sqrt{\frac{\log(2n^2)}{2T_a(t-1)}} \right\}$$

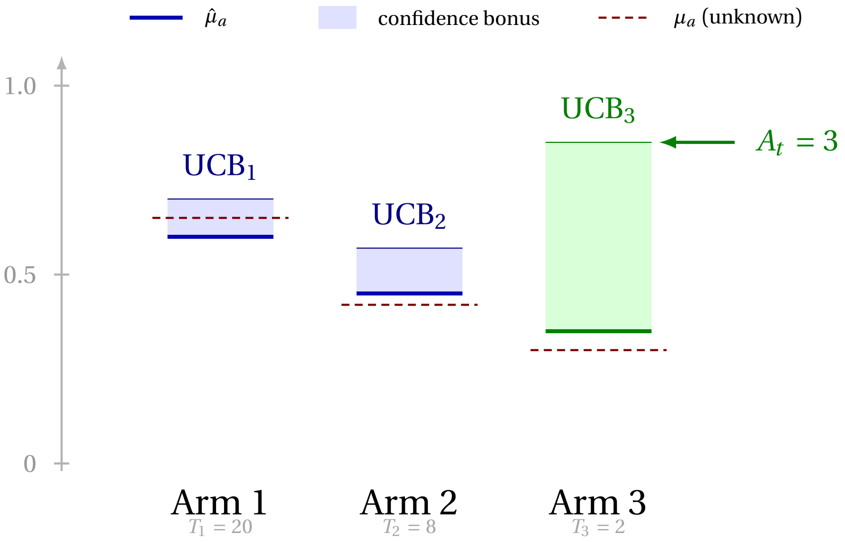
$T_a(t) = \sum_{u=1}^{t} \mathbf{1}_{A_u = a}$ and $\hat{\mu}_a(t) = \frac{1}{T_a(t)} \sum_{u=1}^{t} R_u \mathbf{1}_{A_u = a}$

UCB

$$A_t = \operatorname*{arg\,max}_{a \in 1 \cdots k} \left[\hat{\mu}_a(t-1) + \sqrt{\frac{\log(2n^2)}{2T_a(t-1)}} \right]$$

[Lai, 1987, Katehakis and Robbins, 1995, Auer et al., 2002]

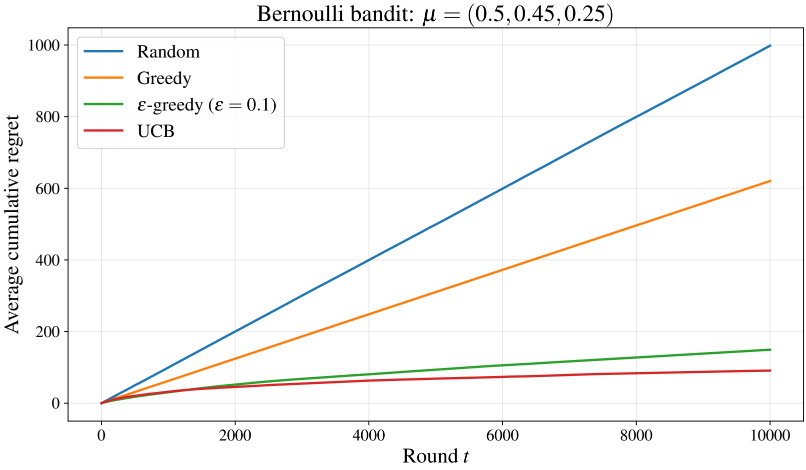
Visualising UCB



Demo

```
python3 ucb_demo.py
```

Regret plot



Bernoulli bandit: $\mu = (0.5, 0.45, 0.25)$
Random
Greedy
ε-greedy ($\varepsilon = 0.1$)
UCB
Average cumulative regret
Round t
0
200
400
600
800
1000
0
2000
4000
6000
8000
10000

Confidence intervals

Theorem (Hoeffding's bound)

Suppose that $X_1, \ldots, X_n$ are $[0,1]$-valued ***independent and identically distributed*** *random variables with mean μ, then for any $\delta \in (0,1)$,*

$$\mathbb{P}\left(|\hat{\mu} - \mu| \geq \sqrt{\frac{\log(2/\delta)}{2n}}\right) \leq \delta$$

where $\hat{\mu} = \frac{1}{n}\sum_{t=1}^{n} X_t$

Applying Hoeffding's bound

	1	2	3	4	5	6	7	8
Arm 1	$.3_{R_1}$	$.7_{R_4}$	$.1_{R_7}$	$.5_{R_9}$	.8	.2	.6	.4
Arm 2	$.9_{R_2}$	$.4_{R_6}$	.6	.1	.3	.7	.5	.8
Arm 3	$.2_{R_3}$	$.5_{R_5}$	$.9_{R_8}$	.3	.7	.1	.4	.6

observed   unobserved

$\hat{\mu}_{4,2} = \frac{1}{4}(0.9 + 0.4 + 0.6 + 0.1) = 0.5$ (mean of first 4 values in Arm 2's row)

Applying Hoeffding's bound

Theorem

With probability at least $1 - n\delta$ *for all rounds* $t \in \{1, \ldots, n\}$ *and arm* a

$$|\hat{\mu}_a(t-1) - \mu_a| \leq \sqrt{\frac{\log(2/\delta)}{2T_a(t-1)}}$$

Proof: $F \subset \bigcup_{t=1}^{n} \left\{ |\hat{\mu}_{t,a} - \mu_a| > \sqrt{\frac{\log(2/\delta)}{2t}} \right\}$

By the **union bound**:

$\mathbb{P}(F) \leq \sum_{t=1}^{n} \mathbb{P}\left(|\hat{\mu}_{t,a} - \mu_a| > \sqrt{\frac{\log(2/\delta)}{2t}} \right) \leq n\delta$

Fundamental identity

The regret can be written in terms of the **gaps** and the expected **play counts**:

$$
\begin{aligned}
\mathbb{E}[\mathrm{Reg}_n] &= \mathbb{E}\left[\sum_{t=1}^{n}(\mu^\star - \mu_{A_t})\right] \\
&= \mathbb{E}\left[\sum_{t=1}^{n}\sum_{a=1}^{k}\mathbf{1}_{A_t=a}(\mu^\star - \mu_a)\right] \\
&= \sum_{a=1}^{k}(\mu^\star - \mu_a)\mathbb{E}\left[\sum_{t=1}^{n}\mathbf{1}_{A_t=a}\right] \\
&= \sum_{a=1}^{k}\Delta_a\mathbb{E}[T_a(n)]
\end{aligned}
$$

Bounding the play counts

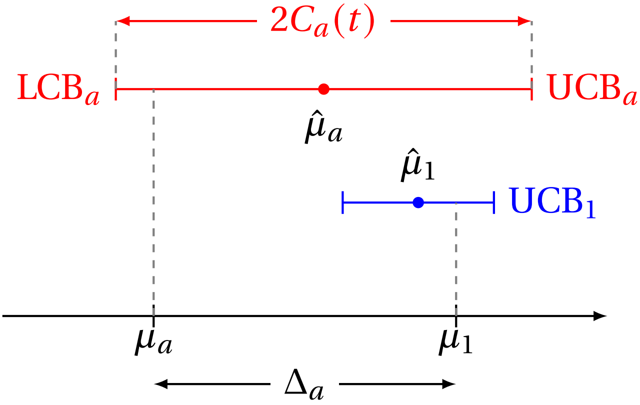
- $\rightarrow$ The regret is $\mathbb{E}[\mathrm{Reg}_n] = \sum_{a=1}^{k} \Delta_a \mathbb{E}[T_a(n)]$
- $\rightarrow$ Need to bound the number of times any suboptimal action a is played
- $\rightarrow$ Let $C_a(t) = \sqrt{\frac{\log(2/\delta)}{2T_a(t-1)}}$. If $A_t = a$ then

$$\mu_a + 2C_a(t) \geq \hat{\mu}_a + C_a(t) \geq \hat{\mu}_1 + C_1(t) \geq \mu_1$$

- $\rightarrow$ Rearranging: $\sqrt{\frac{2\log(2/\delta)}{T_a(t-1)}} = 2C_a(t) \geq \Delta_a$
- $\rightarrow$ $T_a(t-1) \leq \frac{2\log(2/\delta)}{\Delta_a^2}$

Visualising the bound

1. On good event: $\mathrm{LCB}_a \leq \mu_a$ and $\mathrm{UCB}_1 \geq \mu_1$
2. Because a played: $\mathrm{UCB}_a \geq \mathrm{UCB}_1$
3. Therefore $2C_a(t) \geq \Delta_a$



Final bound

- $\rightarrow$ We proved that the bad event has probability at most $nk\delta$
- $\rightarrow$ On the good event $T_a(n) \leq 1 + \frac{2\log(2/\delta)}{\Delta_a^2}$

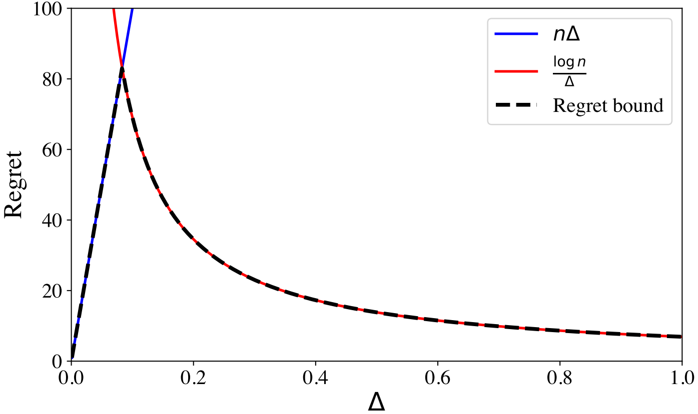
$$\mathbb{E}[\mathrm{Reg}_n] = \mathbb{E}\left[\sum_{a=2}^{k} \Delta_a T_a(n)\right]$$
$$\leq n \cdot nk\delta + \sum_{a=2}^{k} \left(\Delta_a + \frac{2\log(2/\delta)}{\Delta_a}\right)$$

- $\rightarrow$ Choose $\delta = \frac{1}{n^2}$
- $\rightarrow$ $\mathrm{Reg}_n = O\left(\sum_{a=2}^{k} \frac{\log(n)}{\Delta_a}\right)$

Is this good?

- $\rightarrow$ $\text{Reg}_n = O\left(\sum_{a=2}^{k} \frac{\log(n)}{\Delta_a}\right)$
- $\rightarrow$ Let's plot this with $k = 2$

Regret vs Δ



$\longrightarrow$ Worst-case regret occurs when $\Delta \approx \sqrt{\frac{\log n}{n}}$

Minimax bound

Let $\epsilon > 0$

$$
\begin{aligned}
\mathrm{Reg}_n &= \sum_{a=2}^{k} \Delta_a \mathbb{E}[T_n(a)] \\
&\leq n\epsilon + \sum_{a:\Delta_a > \epsilon} \Delta_a \mathbb{E}[T_n(a)] && \left[\textstyle\sum_a T_a(n) \leq n\right] \\
&\leq n\epsilon + \sum_{a:\Delta_a > \epsilon} O\left(\frac{\log(nk)}{\Delta_a}\right) && [\text{prev. bound}] \\
&= O\left(n\epsilon + \frac{k \log(nk)}{\epsilon}\right) && [\Delta_a > \epsilon] \\
&= O\left(\sqrt{nk \log(nk)}\right) && \left[\epsilon = \sqrt{k \log(nk)/n}\right]
\end{aligned}
$$

Theoretical comparison

Let's think about the minimax regret of ϵ-greedy

- $\longrightarrow$ Bandit with means $\mu = (1/2, 1/2 - 1/\sqrt{n\epsilon}, 0)$
- $\longrightarrow$ Regret is at least $n\epsilon$
- $\longrightarrow$ Let $\Delta = 1/\sqrt{n\epsilon}$
- $\longrightarrow$ Learner needs to play the second arm $\sim 1/\Delta^2$ to distinguish first two arms
- $\longrightarrow$ But $1/\Delta^2 = n\epsilon$
- $\longrightarrow$ Regret is also at least $\sqrt{n/\epsilon}$
- $\longrightarrow$ Maximum of the two is $n^{2/3}$

Unknown horizon

→ Simple idea: **doubling trick** – guess some horizon. Run algorithm until the game ends or the guess is hit. Double your guess and restart the algorithm

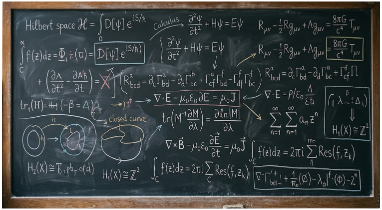
→ Better idea: refine the **confidence level**

$$A_t = \arg\max_{a \in 1 \cdots k} \left(\hat{\mu}_a(t-1) + \sqrt{\frac{\log(2t^2)}{T_a(t-1)}} \right)$$

→ Marginally more technical analysis

Fancy refinements

$$\hat{\mu}_a(t-1) + \sqrt{\frac{1}{2T_a(t-1)} \log}$$



→ $O(\sqrt{nk})$ minimax regret

→ $O\left(\sum_{a \neq a^*} \frac{\log(n)}{\Delta_a}\right)$ problem dependent regret

→ Both optimal

[Audibert and Bubeck, 2009, Degenne and Perchet, 2016, Lattimore, 2018]

Optimality

Both the logarithmic and finite-time bounds are *essentially optimal*

Be careful For every fixed bandit there exists an algorithm that just plays the optimal action

Lower bounds usually have one of two flavours:

→ For any algorithm there exists an instance where the regret is such and such [Nicoló]

→ Any 'good' algorithm cannot be 'too good' on any instance

The noise distributions

- $\rightarrow$ We assumed the rewards were in $[0, 1]$
- $\rightarrow$ Sometimes you know **more**: E.g., the rewards are Bernoulli (in $\{0, 1\}$)
- $\rightarrow$ Sometimes you know **less**:

  "The moments are of the noise distribution are bounded by such and such"

- $\rightarrow$ Or the reward distribution belongs to some parametric family $\mathcal{F} = \{P_\theta : \theta \in \Theta\}$
- $\rightarrow$ The principles stand but the algorithms change. New confidence intervals are needed and the behaviour of the regret can change

Example – Bernoulli rewards

→ Nothing changes except the confidence interval:

$$P_t = \left\{ \nu : \mathrm{KL}(\nu_a, \hat{\mu}_a(t-1)) \leq \frac{\log(\cdot)}{T_a(t-1)} \right\}$$

→ $\mathrm{KL}(p,q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q} \approx \frac{(p-q)^2}{2p(1-p)}$

→ Regret becomes

$$\mathbb{E}[\mathrm{Reg}_n] = O\left(\sum_{a>1} \frac{\Delta_a \log(n)}{\mathrm{KL}(\mu_a, \mu_1)} \right) \approx O\left(\sum_{a>1} \frac{\sigma_a^2 \log(n)}{\Delta_a} \right)$$

[Cappé et al., 2013]

Bayesian viewpoint

- $\rightarrow$ Another viewpoint
- $\rightarrow$ Put prior on the unknown means (and noise model)
- $\rightarrow$ Now *everything is known* and just a big MDP
- $\rightarrow$ Find the optimal policy
- $\rightarrow$ Generally not practical to compute
- $\rightarrow$ Except in one miraculous case [Gittins, 1979]

Beyond finite-armed bandits

→ Add context – rewards for an arm depend on some observable context [Claire]

→ Structured rewards. There many arms but the rewards are connected in some way. E.g., the action set is a convex subset of $\mathbb{R}^d$ and the mean reward is concave

→ Pure exploration

→ Non-stationarity

→ Delayed rewards

→ Changing action sets

→ Privacy

→ And many more

Part II: Policy gradient for stochastic bandits

- → A different motivation
- → How does a classical **RL algorithm** fair in a setting where analysis is possible?
- → Based on lots of recent work: [Mei et al., 2020, 2023, 2024, Baudry et al., 2025, Lattimore, 2026a,b]

Policy gradient dynamics

- → Softmax policy: $\pi_\theta(a) \propto \exp(\theta_a)$
- → Expected reward: $J(\theta) = \sum_{a=1}^{k} \pi_\theta(a)\mu_a$

$$\begin{aligned}
\nabla J(\theta) &= \sum_{a=1}^{k} \mu_a \nabla \pi_\theta(a) \\
&= \sum_{a=1}^{k} \pi_\theta(a)\mu_a \nabla \log \pi_\theta(a) \qquad \text{[log derivative trick]} \\
&= \mathbb{E}_{A\sim\pi_\theta}\left[\mu_A \nabla \log \pi_\theta(A)\right]
\end{aligned}$$

- → Estimate: sample $A_t \sim \pi_{\theta_t}$, observe R_t

$$g_t = R_t \nabla \log \pi_{\theta_t}(A_t) = R_t(e_{A_t} - \pi_{\theta_t})$$

- → $\theta_{t+1} = \theta_t + \eta R_t(e_{A_t} - \pi_{\theta_t})$

Algorithm

1. Initialise $\theta_1 = \mathbf{0}$ and learning rate $\eta > 0$
2. Let $t = 1$
3. Sample $A_t \sim \pi_{\theta_t} \propto \exp(\theta_t)$
4. Observe reward R_t
5. Update $\theta_{t+1} = \theta_t + \eta R_t (e_{A_t} - \pi_{\theta_t})$
6. Go to step 3

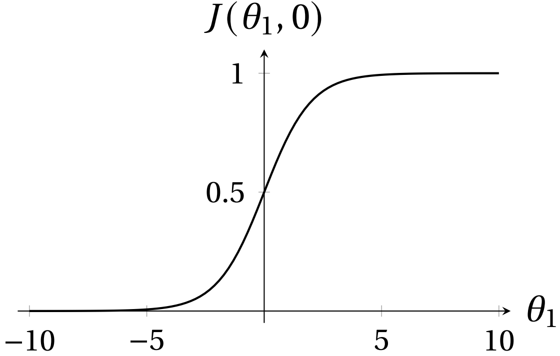
Demo

```
python3 pg_demo.py -eta 0.05 -seed 40
```

```
python3 pg_demo.py -eta 0.5 -seed 40
```

Non-convexity

$$\mu = (1, 0),\ \theta_2 = 0: \quad J(\theta_1, 0) = \frac{e^{\theta_1}}{e^{\theta_1} + e^0} = \frac{1}{1 + e^{-\theta_1}}$$



Non-convexity and flat gradients are kind of worrying

A simple question

*Do we **expect** the probability of the optimal action to increase?*

→ Assume $\mu_1 > \mu_2 \geq \cdots \geq \mu_k$

→ Equivalent to thinking about the expected change of logit differences

$$\nabla J(\theta)_1 - \nabla J(\theta)_a = \pi_a \Delta_a + (\pi_1 - \pi_a)\langle \pi, \Delta \rangle$$

→ The second term only positive if $\pi_1 \geq \pi_a$

So things get hairy if we allow π_1 to fall far below π_a

When $k = 2$ simplifies to $2\pi_a(1 - \pi_a)\Delta_a > 0$

Demo

```
python3 pg_demo.py -eta 0.05 -seed 30 -k=5
```

Master strategy

- $\rightarrow$ Argue that w.h.p. $\pi_{t,1} \gtrsim \pi_{t,a} \; \forall\, t, a$
- $\rightarrow$ We have $\mathbb{E}[\nabla J(\theta)_1] = \pi_1 \langle \pi, \Delta \rangle$
- $\rightarrow$ Then

$$\begin{aligned}
\mathbb{E}[\theta_{n+1,1}] = \eta \mathbb{E}\left[\sum_{t=1}^{n} \nabla J(\theta_t)_1\right] &= \eta \mathbb{E}\left[\sum_{t=1}^{n} \pi_{t,1} \langle \pi_t, \Delta \rangle\right] \\
&\gtrsim \frac{\eta}{k} \mathbb{E}\left[\sum_{t=1}^{n} \langle \pi_t, \Delta \rangle\right] = \frac{\eta}{k} \mathbb{E}[\mathrm{Reg}_n]
\end{aligned}$$

Rearranging shows that $\mathbb{E}[\mathrm{Reg}_n] \leq \frac{k}{\eta} \mathbb{E}[\theta_{n+1,1}]$

Strategy summary

As long as $\pi_{t,1} \gtrsim \pi_{t,a}$ for all t and suboptimal a with high probability, then

$$\mathbb{E}[\mathrm{Reg}_n] \lesssim \frac{k}{\eta}\mathbb{E}[\theta_{n+1,1}]$$

Hence we need to show that

→ $\pi_{t,1} \gtrsim \pi_{t,a}$ indeed holds with high probability

→ That $\mathbb{E}[\theta_{n+1,1}]$ is not too large

Conservation law

The logits satisfy a conservation law:

$$g_t = R_t(e_{A_t} - \pi_{\theta_t})$$

Then

$$\sum_{a=1}^{k} \theta_{t+1,a} = \sum_{a=1}^{k} \theta_{t,a} + \sum_{a=1}^{k} g_{t,a} = \sum_{a=1}^{k} \theta_{t,a}$$

By induction: $\sum_{a=1}^{k} \theta_{t,a} = 0$

Implications of conservation law

$\rightarrow$ $\sum_{a=1}^{k} \theta_{t,a} = 0$

$\rightarrow$ There exists an a such that $\theta_{t,a} \geq 0$

$\rightarrow$ If $\theta_{t,b} \leq -\log(n)$, then

$$\pi_{t,b} = \pi_{t,a} \exp(\theta_{t,b} - \theta_{t,a}) \leq \frac{1}{n}$$

$\rightarrow$ Once the logit is this small, $\theta_{t,b}$ hardly moves

$\rightarrow$ So $\theta_{t,b}$ cannot get much smaller than $-\log(n)$

$\rightarrow$ Consequentially, with high probability

$$\theta_{t,1} = -\sum_{b=2}^{k} \theta_{t,b} \leq (k-1)\log(n)$$

Where are we?

- $\rightarrow$ As long as $\pi_{t,1} \gtrsim \pi_{t,a}$ for all t and a then

$$\mathbb{E}[\mathrm{Reg}_n] \lesssim \frac{k}{\eta}\mathbb{E}[\theta_{n+1,1}]$$

- $\rightarrow$ $\mathbb{E}[\theta_{n+1,1}] = O(k\log(n))$
- $\rightarrow$ How small does the learning rate η need to be to guarantee that $\pi_{t,1} \gtrsim \pi_{t,a}$ w.h.p.?

Signal-to-noise ratio

- $\rightarrow$ Let a be suboptimal and $Z_t = \theta_{t,1} - \theta_{t,a}$
- $\rightarrow$ Want to argue that Z_t stays nearly positive
- $\rightarrow$ A calculation shows that

$$D_t = Z_{t+1} - Z_t = \eta R_t \left[\mathbf{1}_{A_t=1} - \pi_{t,1} - \mathbf{1}_{A_t=a} + \pi_{t,a}\right]$$

- $\rightarrow$ Expectation and variance: when $Z_t \in I$

$$\begin{aligned}
\mathbb{E}_{t-1}[D_t] &= \eta \left[(\pi_{t,1} - \pi_{t,a})r_t + \pi_{t,a}\Delta_a\right] \\
&= \eta \pi_{t,a} \left[(\exp(Z_t) - 1)r_t + \Delta_a\right] \\
&\geq \eta \pi_{t,a}\Delta_a/2 \\
\mathbb{E}_{t-1}[D_t^2] &\leq \eta^2(\pi_{t,1} + \pi_{t,a}) = O(\eta^2 \pi_{t,a})
\end{aligned}$$

Signal-to-noise ratio

With high probability, if $\eta = \tilde{O}(\Delta_a^2)$ then

$$\begin{aligned}
\sum_{t=1}^{\tau} D_t &\gtrsim \sum_{t=1}^{\tau} \mathbb{E}_{t-1}[D_t] - \sqrt{L \sum_{t=1}^{\tau} \mathbb{E}_{t-1}[D_t^2]} \\
&\gtrsim \frac{\eta \Delta_a}{2} \sum_{t=1}^{\tau} \pi_{t,a} - \eta \sqrt{L \sum_{t=1}^{\tau} \pi_{t,a}} \\
&\geq -\frac{\eta L}{2\Delta_a} \geq -\Delta_a
\end{aligned}$$

Final result

Theorem

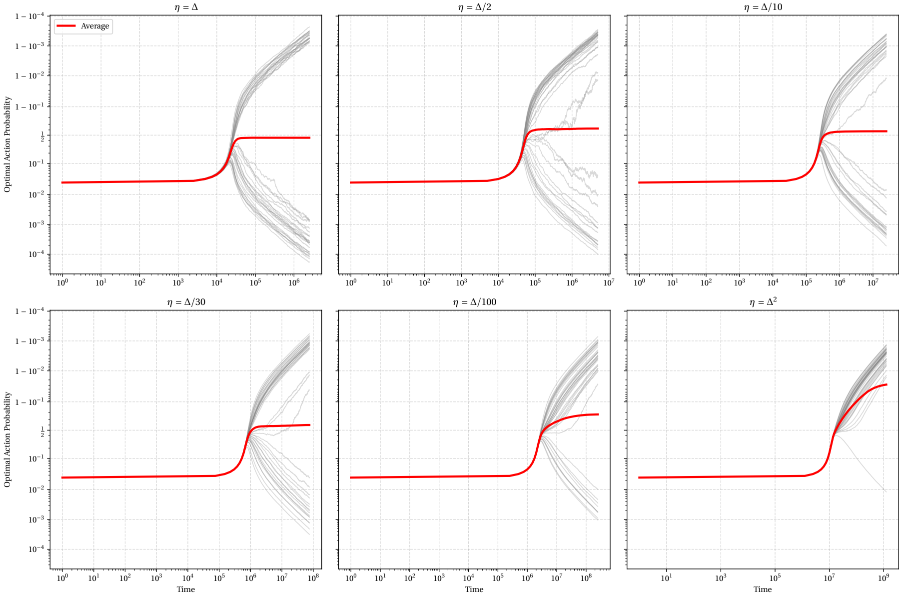
Suppose that $\eta = \tilde{O}(\Delta_{\min}^2)$ *then*

$$\mathbb{E}[Reg_n] = O\left(\frac{k^2 \log(n)}{\eta}\right)$$

Not great compared to UCB:

$$\mathbb{E}[\mathrm{Reg}_n] = O\left(\sum_{a>1} \frac{\log(n)}{\Delta_a}\right)$$

Experiments



What’s next?

$\rightarrow$ Adversarial bandits – Nicoló

$\rightarrow$ Structured/contextual bandits – Claire

Questions and references

J.-Y. Audibert and S. Bubeck. Minimax policies for adversarial and stochastic bandits. In *Proceedings of the 22nd Conference on Learning Theory*, pages 217–226, 2009.

P. Auer, N. Cesa-Bianchi, and P. Fischer. Finite-time analysis of the multiarmed bandit problem. *Machine Learning*, 47: 235–256, 2002.

D. Baudry, E. Johnson, S. Vary, C. Pike-Burke, and P. Rebeschini. Does stochastic gradient really succeed for bandits? In *NeurIPS*, 2025.

O. Cappé, A. Garivier, O. Maillard, R. Munos, and G. Stoltz. Kullback–Leibler upper confidence bounds for optimal sequential allocation. *The Annals of Statistics*, 41(3):1516–1541, 2013.

R. Degenne and V. Perchet. Anytime optimal algorithms in stochastic multi-armed bandits. In *Proceedings of the 33rd International Conference on Machine Learning*, pages 1587–1595, New York, NY, USA, 20–22 Jun 2016. JMLR.org.

J. Gittins. Bandit processes and dynamic allocation indices. *Journal of the Royal Statistical Society. Series B (Methodological)*, 41(2):148–177, 1979.

M. N. Katehakis and H. Robbins. Sequential choice from several populations. *Proceedings of the National Academy of Sciences of the United States of America*, 92(19):8584, 1995.

T. L. Lai. Adaptive treatment allocation and the multi-armed bandit problem. *The Annals of Statistics*, pages 1091–1114, 1987.

T. Lattimore. Refining the confidence level for optimistic bandit strategies. *Journal of Machine Learning Research*, 2018.

T. Lattimore. A diffusion analysis of policy gradient for stochastic bandits. *arXiv preprint arXiv:2603.10219*, 2026a.

T. Lattimore. A lyapunov analysis of softmax policy gradient for stochastic bandits. *arXiv preprint arXiv:2603.26547*, 2026b.

J. Mei, C. Xiao, C. Szepesvári, and D. Schuurmans. On the global convergence rates of softmax policy gradient methods. In *ICML*, 2020.

J. Mei, Z. Zhong, B. Dai, A. Agarwal, C. Szepesvári, and D. Schuurmans. Stochastic gradient succeeds for bandits. In *ICML*, 2023.

J. Mei, B. Dai, A. Agarwal, S. Vaswani, A. Raj, C. Szepesvári, and D. Schuurmans. Small steps no more: Global convergence of stochastic gradient bandits for arbitrary learning rates. In *NeurIPS*, 2024.

W. Thompson. On the likelihood that one unknown probability exceeds another in view of the evidence of two samples. *Biometrika*, 25(3/4):285–294, 1933.